

Context

A tip is considered as a monetary incentive that is given by the customers or guests for polite, prompt, and efficient service provided by the staff. The practice of giving tips has been continuing for a very long time and has become particularly popular in service industries such as hotels and restaurants.

Objective

Chef's Kitchen is one of the most popular restaurants in the city of San Diego and acts as a one-stop destination for food lovers. The polite and efficient service provided by the restaurant staff often gets them tips from the customer. As a Data Analyst for the restaurant, you have been asked to analyze the data provided to identify the patterns and trends in the revenue and tips received from customers across different demographics and come up with informative visualizations to convey the insights obtained from the analysis.

The dataset consists of the following features:

- **order_id**: unique identifier of each order
- **day**: day of the week on which the customer visited the restaurant (Thur/Fri/Sat/Sun)
- **time**: time of the day at which the customer visited the restaurant (Lunch or Dinner)
- **size**: number of people present at the table
- **smoker**: whether the table included smokers (No: The table did not include smokers, Yes: The table included smokers)
- **sex**: gender of the payer of the bill (Male or Female)
- **total_bill**: the bill amount in dollars
- **tip**: the tip amount in dollars

Key Steps to be performed -

1. Import all the necessary packages and load the dataset
2. Get a brief overview of the dataset
3. Explore the dataset to come up with meaningful insights